

PIYUSH GARG

Delhi NCR | erpiyushgarg17@gmail.com | GitHub | LinkedIn | Portfolio

Education

Bachelor of Technology in AI & Data Science

Akhilesh Das Gupta Institute of Professional Studies

Expected Graduation: May 2028

Current CGPA: 7.8/10.0

Relevant Coursework: Deep Learning, Data Structures & Algorithms, Calculus, Probability

Central Board of Secondary Education (CBSE 12th)

Bal Bhavan International School

Graduated: June 2024

Score: 75%

Skills

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, Generative Models (GANs, Diffusion, Autoencoders)

Programming: Python, C++, C, JavaScript

Backend & AI Systems: FastAPI

Scientific Computing: NumPy, Pandas, Matplotlib

Languages: English, Hindi

Projects

ToxiFlow — Molecular Toxicity Prediction with ChemBERTa

GitHub

- Implemented a modular Transformer-based pipeline for molecular toxicity prediction using SMILES representations and ChemBERTa with PyTorch.
- Developed data preprocessing and tokenization workflows for molecular sequences, including class-imbalance handling for robust toxicity classification.
- Implemented ChemBERTa-based feature extraction with mean pooling to generate molecular representations for downstream toxicity prediction.
- Designed the pipeline for extensibility across molecular property prediction and drug-discovery tasks through modular preprocessing, modeling, and evaluation components.

SynapseAI — Git-Based Repository Memory System (V3 in Progress)

GitHub

- Developing a local-first Repository Intelligence Engine that incrementally transforms Git changes into structured repository knowledge and a persistent knowledge graph for scalable codebase understanding.
- Implementing incremental repository analysis using Git change detection and AST-based parsing to index files, classes, functions, imports, calls, dependencies, and symbol relationships.
- Designing graph-based retrieval and semantic indexing to identify code relevant to user queries while decoupling LLM context requirements from overall repository size.
- Developing a derived knowledge pipeline that generates architectural summaries from structured repository knowledge while maintaining the repository index as the canonical source of truth for downstream AI reasoning.

Diffusion-Based Generative Modeling for Sparse Detector Data

GitHub — Paper

- Implemented a denoising diffusion probabilistic model (DDPM) pipeline in PyTorch for sparse calorimeter detector event generation.
- Designed timestep-conditioned denoising workflows, data preprocessing pipelines, and physics-aware evaluation strategies for generative modeling of detector showers.
- Explored sparse detector representations, diffusion sampling techniques, and high-dimensional data generation under constrained computational settings.
- Investigated architectural and computational bottlenecks affecting training stability, reconstruction fidelity, and scalability of diffusion-based learning systems.